Coherent Sequence Unified Coupled-Field Theory Linking Macro-Cosmology and Micro-Vacuum Dynamics

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Abstract

To elucidate the microscopic origin and macroscopic structure of the universe and to resolve the Coincidence Problem-specifically, "why the energy densities of dark energy and dark matter have become comparable in the present cosmic epoch"-i formulate a hypothetical coupled-field system termed the **Coherent Sequence Unified Field Equations**. This framework integrates the mass gravity of dark matter, the expansion-driving force of dark energy, and the wave dynamics of Higgs fields and particles into a unified formalism. The formulation is underpinned by the mathematical isomorphism shared by the background-fluctuation decomposition structures of macroscopic dark energy fields and microscopic Higgs vacua, combining an interacting dark energy-dark matter wave equation with a Higgs-fermion coupling measurement equation into a single theoretical system.

Definition of Coherent Sequence:

A "sequence" refers to a macroscopic phase propagation regime wherein spatial randomness is eliminated, and fields propagate with a well-defined frequency ω and wave vector $\mathbf{k}$ under phase and amplitude coherence. This denotes a state where the local phase $\theta(\mathbf{x},t)$ of the field maintains a consistent dynamical relationship across space and time. This state is characterized by relations among phase, amplitude, frequency, and wave number, and the degree of phase coherence is quantified by a dedicated coherence function: $\psi(\mathbf{x},t) = A(\mathbf{x},t)e^{i\theta(\mathbf{x},t)}, \quad \theta(\mathbf{x},t) = \mathbf{k}\cdot\mathbf{x} - \omega t + \theta_0$

where A denotes the amplitude, θ denotes the phase, $\mathbf{k}$ denotes the wave vector, and ω denotes the angular frequency. The quantification of phase coherence is given by:$C_{\text{seq}} = \left|\left\langle e^{i\theta} \right\rangle\right|, \quad 0 \leq C_{\text{seq}} \leq 1$

Unified Field Theory Framework:

The macroscopic cosmological field (Φ) and the microscopic vacuum Higgs field (ϕ) both share a common variational principle known as "Background-Fluctuation Isomorphism," and they are bound into a unified coupled system via Yukawa couplings and energy-momentum transfer (Q^{ν}).

Macro-Micro Isomorphism:

Going beyond a mere morphological decomposition of fields, the macroscopic dark energy fluctuation $\psi(\mathbf{x}, t)$ and the microscopic Higgs fluctuation $H(x)$ share identical nonlinear Mexican-hat potential curvature structures and vacuum expectation value modulation mechanisms, serving as the basis for their mathematical isomorphism.

Coincidence Problem Consistency:

The energy exchange initiated by the interaction term Q^{ν} maps one-to-one to the matter coupling term $\beta\rho_m$ in the subsequent wave equations.

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Main Text

A. Foundational Hypotheses and Derived Wave Equations

Cosmological equations unify the dimensions of density and pressure, while field equations ensure dimensional consistency across all terms:

$$S_{\text{total}} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G}(R - 2\Lambda) + \mathcal{L}_{\text{DE}} + \mathcal{L}_{\text{DM}} + \mathcal{L}_{\text{H}} + \mathcal{L}_{\text{f}} + \mathcal{L}_{\text{int}} \right]$$

where each Lagrangian corresponds to dark energy, dark matter, the Higgs field, fermions, and their mutual interactions:

$$\mathcal{L}_{\text{int}} = \mathcal{L}_{\text{int,DE-DM}}(\Phi, \rho_m) + \mathcal{L}_{\text{int,Higgs-Fermion}}(h, \psi_f) + \mathcal{L}_{\text{int,Macro-Micro}}(\delta\Phi, h) + \mathcal{L}_{\text{int,Probe}}$$

This expression assumes standard spacetime coordinate conventions such as $x^0 = ct$. The interaction Lagrangian density $\mathcal{L}_{\text{int}}$ mediates the physical bridging and internal coupling between the macroscopic cosmological domain (dark energy-dark matter) and the microscopic vacuum domain (Higgs-fermion) addressed in this work.

$\mathcal{L}_{\text{int,DE-DM}}$ **(Macroscopic Interaction):** Coupling between the dark energy field and dark matter. To bridge the Coincidence Problem and energy-momentum conservation (IDE model), this interaction term enables direct energy exchange between the dark energy field Φ (or its perturbation $\delta\Phi$) and the dark matter component.

Coupled Fields: Macroscopic dark energy field Φ (or fluctuation ψ) and dark matter density/fluid field ρ_m (or dark matter field).

Physical Role: Acts as a direct source term ($\beta\rho_m$) in regions of high dark matter concentration (such as galaxy clusters) on the phase sequence and effective mass of the dark energy field, dynamically adjusting the energy density ratio of the two components to be comparable in the current epoch.

$\mathcal{L}_{\text{int,Higgs-Fermion}}$ **(Microscopic Interaction):** Yukawa coupling between the Higgs field and fermions. Describing the microscopic vacuum dynamics of the Standard Model, this governs mass generation and dynamic interactions between the Higgs doublet ϕ (or Higgs fluctuation field $h(x)$) and the fermion field ψ_f.

Coupled Fields: Higgs field fluctuation $h(x)$ (or vacuum expectation value v_{EW}) and fermion spinor fields $\psi_f, \bar{\psi}_f$.

Physical Role: Simultaneously induces static mass generation terms and dynamic Higgs

fluctuation wave terms, mediating phase modulation and resonant responses driven by external probes such as terahertz radiation.

$\mathcal{L}_{\text{int,Macro}-\text{Micro}}$ **(Macroscopic-Microscopic Unified Bridge):** Isomorphic coupling between the dark energy field and the Higgs vacuum. This coupling term connects the core hypothesis of this work—the mathematical isomorphism between the background-fluctuation decomposition structures of the macroscopic dark energy field and the microscopic Higgs vacuum—directly at the Action level.

Coupled Fields: Macroscopic dark energy background field $\Phi_0(t)$ / perturbation $\delta\Phi$ and microscopic Higgs vacuum v_{EW} / fluctuation field $h(x)$.

Physical Role: Numerically constrains the local resonant background effects exerted by macroscopic cosmological expansion and phase propagation sequences onto the energy scale of microscopic vacuum Higgs fluctuations.

$\mathcal{L}_{\text{int,Probe}}$ **(External Probe/Measurement Source Coupling):** Interaction with external driving pulses (e.g., terahertz waves, high-frequency electromagnetic fields) examined in theoretical verification and measurement equations (A.3 and B.1.4).

Coupled Fields: External gauge/probe field A_μ^{probe} (or source current) and matter/field perturbations.

Physical Role: Derives observational response functions ($\mathcal{S}_{\text{meas}}, \mathcal{S}_{\text{drive}}$) for experimentally measuring the phase modulation of coherent sequences and the interference patterns of fermion matter waves.

Common Structure Separating Background and Fluctuations for Both Fields:

$$\Phi = \bar{\Phi} + \delta\Phi, \quad \phi = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_{\text{EW}} + h \end{pmatrix}$$

Structural Correspondence of Background-Fluctuation Decomposition:

$$\mathcal{M}: (\bar{\Phi}, \delta\Phi, \mathcal{D}_{\text{DE}}) \longrightarrow (v_{\text{EW}}, h, \mathcal{D}_{\text{H}})$$

Consistency with the Coincidence Problem (Energy Exchange between Dark Matter and Dark Energy):

$$\nabla_\mu T_{\text{de}}^{\mu\nu} = Q^\nu, \quad \nabla_\mu T_{\text{m}}^{\mu\nu} = -Q^\nu$$

$\beta\rho_m$: Source term of the field driven by dark matter density.

$\mathcal{S}_{\text{seq}}[\delta\Phi]$: Nonlinear self-interaction term of the field itself.

Q^ν : Four-vector representing energy-momentum transfer between the two components.

Nomenclature:

Φ : Macroscopic dark energy field, $\bar{\Phi}(t)$: Background of the macroscopic dark energy field, $\delta\Phi(\mathrm{x},t)$: Perturbation, $H_{\mathrm{Hub}}(t)$: Hubble parameter, $h(x)$: Higgs particle fluctuation field, v_{EW} : Higgs vacuum expectation value, h : Higgs field fluctuation.

1. Coincidence Problem

As the universe expands, dark matter dilutes while dark energy remains approximately constant. To explain the coincidence puzzle of why the abundances of these two components have become comparable precisely in the current cosmic era—rather than being a mere accident—coupling terms that allow energy exchange between the two components are introduced into the conservation equations of the energy-momentum tensor. We explicitly emphasize that the four-vector exchange term Q^ν in the macroscopic energy-momentum conservation equation and the scalar field source term $\beta\rho_m$ are not independent phenomenological assumptions. Instead, they are two sides of the same physical interaction, derived respectively via Euler-Lagrange variation and covariant divergence of the energy-momentum tensor from a common interaction action integral $\mathcal{S}_{\mathrm{seq}}$. When describing the same dark energy-dark matter interaction, both terms must be derived from a common interaction action. This ensures mathematical consistency between the dynamics of the dark energy field and the energy-momentum exchange between the two components. (Note that while $\beta\rho_{\mathrm{DM}}$ can alternatively be designated as an independent phenomenological source, in such cases the two terms do not connect automatically unless a specific relation with Q^ν is separately postulated.)

$$\nabla_\mu T^{\mu\nu}_{(\mathrm{de})} = Q^\nu, \quad \nabla_\mu T^{\mu\nu}_{(\mathrm{m})} = -Q^\nu, \ \nabla_\mu \left(T^{\mu\nu}_{(\mathrm{de})} + T^{\mu\nu}_{(\mathrm{m})}\right) = 0$$

Although the energy-momentum of each individual component can be exchanged, their sum is conserved. Such a structure characterizes interacting dark matter-dark energy models.

Relation between Q^ν and $\beta\rho_{\mathrm{DM}}$: $\nabla_\mu T^{\mu\nu}_{\mathrm{DE}} = Q^\nu, \quad \nabla_\mu T^{\mu\nu}_{\mathrm{DM}} = -Q^\nu$

(Q^ν : Four-vector representing the energy-momentum transfer between dark energy and dark matter)

($\beta\rho_{\mathrm{DM}}$: Scalar source term entering the wave equation of dark energy perturbations)

2. Interacting Dark Energy (IDE)

Definition of the Common Interaction Term: We assume that the dark energy field and dark matter share a common interaction term: $\mathcal{L}_{\mathrm{int}} = \mathcal{L}_{\mathrm{int}}(\Phi_{\mathrm{DE}}, \mathrm{DM})$

By incorporating this interaction term into the total action and subsequently computing the Euler-Lagrange equations for the dark energy field alongside the covariant divergence of the energy-momentum tensor for dark matter, we can construct the framework such that the source term of the dark energy field equation and the energy-momentum transfer vector originate from the same underlying interaction:

$\mathcal{D}_{\mathrm{DE}}\delta\Phi = \mathcal{S}_{\mathrm{seq}} + \beta\rho_{\mathrm{DM}}, \quad \nabla_\mu T^{\mu\nu}_{\mathrm{DE}} = Q^\nu, \quad \nabla_\mu T^{\mu\nu}_{\mathrm{DM}} = -Q^\nu$

Here, β denotes the coupling constant of the scalar field equation, and Q^ν represents the energy-momentum transfer vector. When derived from a single interaction term, the precise relation between the two terms is determined by the chosen action and sign conventions.

2.1. Dark Matter: Foundations of Existing Equations

2.1.1. Galactic Rotation Velocity Equation: $v_{\mathrm{rot}}(r) = \sqrt{\frac{GM_{\mathrm{tot}}(<r)}{r}}$

This expression is applied under the assumptions of spherically symmetric mass distribution and circular orbits, where:

$M_{\mathrm{tot}}(<r) = M_{\mathrm{b}}(<r) + M_{\mathrm{DM}}(<r) + M_{\mathrm{gas}}(<r) + \cdots$

($M_{\mathrm{tot}}(<r)$ is the enclosed total mass within radius r, while M_{b}, M_{DM}, and M_{gas} represent the contributions from baryonic matter, dark matter, and gas, respectively.)

2.1.2. Density Evolution in an Expanding Universe (Contribution within Friedmann Equations): The standard non-interacting dilution relation $\rho_m(a) \propto \frac{1}{a^3}$ is generalized by the continuity equation: $\dot{\rho}_m + 3H_{\mathrm{Hub}}\rho_m = Q_m$

The formal solution is given by: $\rho_m(t) = a(t)^{-3}\left[\rho_{m0} + \int_{t_0}^{t} a\,(t')^3 Q_m(t')\,dt'\right], \quad a(t_0) = 1$

If $Q_m = 0$, the conventional a^{-3} dilution law is recovered, whereas $Q_m \neq 0$ modifies the density evolution according to the interaction.

2.1.3. Einstein Field Equations (Including the Cosmological Constant):

Incorporating the dynamical dark energy field into the energy-momentum tensor yields:

$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mathrm{DM}\,\mu\nu} + T_{\mathrm{DE}\,\mu\nu} + T_{\mathrm{other}\,\mu\nu}\right)$

where Λ is either not separately maintained or treated as a distinct residual cosmological constant component: $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$

2.1.4. Equation of State:

$w = \frac{p}{\varepsilon}, \quad [\varepsilon] = [p] = \mathrm{J\,m^{-3}}$ (where ε denotes the energy density and p denotes the pressure.)

2.1.5. Cosmological Constant Model (Determining Accelerated Cosmic Expansion):

Friedmann Acceleration Equation: $\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\varepsilon + 3p)$ under natural units $c = \hbar = 1$.

2.2. Dark Energy:

2.2.1. Basic Assumptions and Field Setup: The dark energy field $\Phi(x^\mu)$ is a macroscopic scalar field defined in spacetime coordinates $x^\mu = (t, \mathbf{x})$. Rather than exhibiting abrupt fluctuations (high-frequency oscillations at the particle level), it possesses a smooth sequential nature, undulating with a gradual and regular phase progression on a cosmological scale. In other words, the field expansion is represented by the combination of a background value $\bar{\Phi}$ and a smooth wave fluctuation $\psi(\mathbf{x}, t)$:

$\Phi(\mathbf{x}, t) = \bar{\Phi}(t) + \delta\Phi(\mathbf{x}, t)$

2.2.2. Construction of the Lagrangian Density:

To describe the undulating motion of the dark energy field propagating through spacetime, a coherence sequence term is added to the standard quintessence Lagrangian:

$\mathcal{L} = -\frac{1}{2}g^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi - V(\Phi) + \frac{\lambda_{\mathrm{seq}}}{4!}\left(\frac{\Phi}{\Phi_{\mathrm{pl}}}\right)^2\left(\partial_\mu\Phi\partial^\mu\Phi\right)$

$K(\Phi) = 1 - \frac{\lambda_{\mathrm{seq}}}{12}\left(\frac{\Phi}{\Phi_{\mathrm{pl}}}\right)^2$

$\mathcal{L} = -\frac{1}{2}K(\Phi)\partial_\mu\Phi\,\partial^\mu\Phi - V(\Phi)$

$\nabla_\mu(K(\Phi)\nabla^\mu\Phi) - \frac{1}{2}K'(\Phi)\partial_\mu\Phi\partial^\mu\Phi - V'(\Phi) = 0 \quad \left(K'(\Phi) = \frac{dK}{d\Phi},\ V'(\Phi) = \frac{dV}{d\Phi}\right)$

$\lambda_{\mathrm{seq}} \neq \lambda_{\mathrm{H}}$

$\mathcal{S}_{\mathrm{seq}} = \mathcal{S}_{\mathrm{nl}} + \mathcal{S}_{\mathrm{drive}}$

Applying rigorous Euler-Lagrange variation to the non-canonical kinetic term $K(\Phi) = 1 - \frac{\lambda_{\text{seq}}}{12}\left(\frac{\Phi}{\Phi_{\text{pl}}}\right)^2$ introduced in this Lagrangian yields an additional derivative-derived term, $\nabla_\mu K(\Phi)\nabla^\mu\Phi$, via the product rule. These higher-order derivative terms are not independent sources in the perturbation equation derivation process; rather, they are entirely absorbed as constituent components of the nonlinear self-interaction source term $\mathcal{S}_{\text{seq}}[\psi]$.

2.2.3. Derivation of Field Equations in Friedmann-Robertson-Walker (FRW) Spacetime:

Applying the Euler-Lagrange equation under a dynamical metric incorporating cosmic expansion (scale factor $a(t)$) yields a generalized Klein-Gordon-type wave equation reflecting spacetime curvature and the expansion rate of the universe:

$$\ddot{\Phi} + 3H_{\text{Hub}}\dot{\Phi} - \frac{c^2}{a^2}\nabla^2\Phi + V'(\Phi) = J_{\text{DE}}$$

Decomposing the total field as $\Phi = \bar{\Phi} + \delta\Phi$, the background equation is given by:

$$\ddot{\bar{\Phi}} + 3H_{\text{Hub}}\dot{\bar{\Phi}} + V'(\bar{\Phi}) = \bar{J}_{\text{DE}}$$

Subtracting the background equation from the total field equation yields the fluctuation equation: $\delta\ddot{\Phi} + 3H_{\text{Hub}}\delta\dot{\Phi} - \frac{c^2}{a^2}\nabla^2\delta\Phi + V''(\bar{\Phi})\delta\Phi = \delta J_{\text{DE}} - \mathcal{N}(\delta\Phi)$

Nonlinear residual term: $\mathcal{N}(\delta\Phi) = V'(\bar{\Phi} + \delta\Phi) - V'(\bar{\Phi}) - V''(\bar{\Phi})\delta\Phi$

Expansion for small perturbations: $\mathcal{N}(\delta\Phi) = \frac{1}{2}V'''(\bar{\Phi})(\delta\Phi)^2 + \mathcal{O}((\delta\Phi)^3)$

Defining the effective mass squared as $m_{\text{eff}}^2(\bar{\Phi}) = V''(\bar{\Phi})$ directly connects the mass term of the linear fluctuation equation to the curvature of the potential.

2.3. Dark Energy-Dark Matter Wave Equation: Coherent Sequence Wave Equation

In this study, the dark matter coupling model, the self-interaction model, and the coherent sequence model are distinguished sequentially. The first model serves as a reference model including driving by matter density; the second model is a special model isolating nonlinear self-interaction. The final unified model incorporates self-interaction, phase driving, and dark matter coupling into a single fluctuation equation.

2.3.1. General Form of the Wave Equation Used in Dark Matter and Dark Energy Extensions:

$$\left[\frac{\partial^2}{\partial t^2} + 3H(t)\frac{\partial}{\partial t} - \frac{c^2}{a(t)^2}\nabla^2 + m_{\text{eff}}^2(\delta\Phi)\right]\psi(\mathbf{x}, t) = \beta\rho_m(\mathbf{x}, t)$$

$\mathcal{D}_{\mathrm{FLRW}}\psi = \beta\rho_m, \quad \langle\delta\Phi\rangle = 0$

Here, the background and perturbations are not independently prescribed variables, but rather two components resulting from the decomposition of a single total field.

$\Phi(\mathbf{x},t) = \bar{\Phi}(t) + \delta\Phi(\mathbf{x},t)$:

Background scalar field, the average component governing the macroscopic background density and equation of state of dark energy, chosen as a cosmological background solution. In a homogeneous FRW background, it does not depend on spatial coordinates.

$\delta\Phi(\mathbf{x},t)$: Local perturbation from the background solution.

$\psi(\mathbf{x},t)$: Fluctuation wave function, a dynamical perturbation component mediating interactions and phase modulation between dark energy and dark matter in local domains.

∂ : Partial differential symbol.

$\frac{\partial^2\psi}{\partial t^2}$: Standard second-order derivative term describing the temporal evolution of the field.

$3H\frac{\partial\psi}{\partial t}$: Hubble friction term appearing in an expanding FRW universe.

$-\frac{c^2}{a^2}\nabla^2\psi$: Spatial propagation term reflecting cosmic expansion.

$m_{\mathrm{eff}}^2(\Phi)\psi$: Effective mass or potential-derived term.

$\beta\rho_m$: Source term directly coupling matter density to the field equation.

Step 1: Matter-Coupled Reference Model: Reference model including driving by dark matter density.

$\mathcal{D}_0\delta\Phi = \beta\rho_m, \quad \mathcal{D}_0 = \frac{\partial^2}{\partial t^2} + 3H_{\mathrm{Hub}}\frac{\partial}{\partial t} - \frac{c^2}{a^2}\nabla^2 + m_{\mathrm{eff}}^2(\bar{\Phi})$

($\mathcal{D}_0$ is the linear FRW field operator derived from the background field.)

2.3.2. Extended Equation Differentiating Self-Interaction and Matter Coupling:

$\left[\frac{\partial^2}{\partial t^2} + 3H_{\mathrm{Hub}}(t)\frac{\partial}{\partial t} - \frac{c^2}{a(t)^2}\nabla^2 + m_{\mathrm{eff}}^2(\bar{\Phi})\right]\delta\Phi(\mathbf{x},t) = \mathcal{S}_{\mathrm{seq}}[\delta\Phi] + \beta\rho_m(\mathbf{x},t)$

Here, the effective mass squared $m_{\mathrm{eff}}^2(\bar{\Phi})$ reflecting the dynamical effects of the non-canonical kinetic term $K(\bar{\Phi})$ on the background field $\bar{\Phi}(t)$ is defined as follows:

$m_{\mathrm{eff}}^2(\bar{\Phi}) = \frac{1}{K(\bar{\Phi})}\left[V''(\bar{\Phi}) - \frac{1}{2}K''(\bar{\Phi})\dot{\bar{\Phi}}^2 - \frac{d}{dt}\left(\frac{3H_{\mathrm{Hub}}K\prime(\bar{\Phi})\dot{\bar{\Phi}}}{K(\bar{\Phi})}\right) + \cdots\right]$

Substituting these relations into the potential $V(\Phi)$ or the effective mass term $m_{\text{eff}}^2(\Phi)$ naturally separates the contribution from the background field and the nonlinear self-interaction term from the fluctuation field. Here, the zeroth-order term $m_{\text{eff}}^2(\overline{\Phi})$ acts as the linear effective mass term of the wave equation, while higher-order terms are absorbed into the self-interaction source term $\mathcal{S}_{\text{seq}}[\delta\Phi]$.

$\mathcal{S}_{\text{seq}}[\delta\Phi]$: Self-interaction source term proposed to represent internal self-interaction or sequence formation mechanisms within the dark energy field.

$\beta\rho_m(\mathbf{x},t)$: Matter coupling term assuming that the dark matter density influences the evolution of the dark energy field (provided that natural units $c=1,\hbar=1$ are adopted or the dimension of β is explicitly specified).

$$\left[\frac{\partial^2}{\partial t^2}\right]=\mathrm{T}^{-2},\quad \left[3H\frac{\partial}{\partial t}\right]=(\mathrm{T}^{-1})\cdot(\mathrm{T}^{-1})=\mathrm{T}^{-2},\quad \left[\frac{c^2}{a^2}\nabla^2\right]=\left[\frac{c^2}{a(t)^2}\nabla^2\right]=(\mathrm{LT}^{-1})^2\cdot\mathrm{L}^{-2}=\mathrm{T}^{-2}$$

Assuming $a(t)$ is defined as dimensionless and ∇ is a spatial derivative with respect to length, the effective mass term must also share the same dimension as the other operators in the equation (given that the spatial derivative operator ∇ with respect to coordinate distance x has units of L^{-1}).

Effective mass term on the LHS: Denoted as $m_{\text{eff}}^2(\overline{\Phi})$, representing a constant frequency scale tied to the background.

Source terms on the RHS: Clearly separates the self-interaction of the fluctuation field $\mathcal{S}_{\text{seq}}[\delta\Phi]$ from the dark matter density coupling term $\beta\rho_m(\mathbf{x},t)$.

$\left[m_{\text{eff}}^2\right]=\mathrm{T}^{-2}$: Defining m_{eff} not as mass itself but as an effective angular frequency, or using the actual mass M_{eff}, the term can be expressed as $\frac{M_{\text{eff}}^2c^4}{\hbar^2}\delta\Phi$, satisfying $[\beta\rho_m]=[\delta\Phi]\,\mathrm{T}^{-2}$. Here, the dimension of β depends on the definition of ρ_m and $\delta\Phi$. Therefore, under natural units ($c=1,\hbar=1$), the common dimension of the derivative operators and the effective mass term on the LHS is T^{-2}. When the dark matter density ρ_m has the energy density dimension of $\mathrm{T}^{-2}\mathrm{L}^{-2}$ (or an equivalent scale), the coupling factor is rigorously established as a dimensionless or appropriately dimensioned scaling factor to precisely satisfy the dimensional balance of T^{-2} across the entire equation.

Step 2: Self-Interaction Extended Model: A specialized model isolating matter coupling, studied independently of matter coupling. $\mathcal{D}_0\delta\Phi=\mathcal{S}_{\text{nl}}[\delta\Phi]$ (corresponding to the case where $\beta=0$). $[\mathcal{D}_0\delta\Phi]=[\beta\rho_{\text{DM}}],\quad [\beta]=\frac{[\delta\Phi]\,\mathrm{s}^{-2}}{[\rho_{\text{DM}}]}$

Using standard canonical normalization for a scalar field in 4-dimensional spacetime and

defining ρ_{DM} as energy density in natural units $c = \hbar = 1$, we have $[\delta\Phi] = \mathrm{M}$, $[\mathcal{D}_0] = \mathrm{M}^2$, and $[\rho_{\mathrm{DM}}] = \mathrm{M}^4$, leading to $[\beta] = \mathrm{M}^{-1}$. In this specific setup, β is not a dimensionless constant; changing the field normalization or source definition alters the dimension of the coupling constant.

2.3.3. Coherent Sequence Wave Equation:

Incorporating a nonlinear phase modulation source term $\mathcal{S}_{\mathrm{seq}}$—which controls the smooth and steady undulations (sequences) characteristic of dark energy alongside the cosmological expansion friction term—yields the final wave equation:

$$\left[\frac{\partial^2}{\partial t^2} + 3H(t)\frac{\partial}{\partial t} - \frac{c^2}{a(t)^2}\nabla^2 + m_{\mathrm{eff}}^2(\Phi)\right]\psi(\mathbf{x},t) = \mathcal{S}_{\mathrm{drive}}(\mathbf{x},t) = \mathrm{Re}\left[\gamma_{\mathrm{seq}}\left(\nabla\cdot\mathbf{v}_{\mathrm{seq}}\right)e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}\right]$$

$$\left[\nabla\cdot\mathbf{v}_{\mathrm{seq}}\right] = \mathrm{T}^{-1}, \quad \left[\gamma_{\mathrm{seq}}\right] = [\delta\Phi]\,\mathrm{T}^{-1}$$

$\ddot{\Phi} + 3H(t)\dot{\Phi}$: Temporal evolution and Hubble friction term; indicates how the fluctuation energy of the dark energy field is diluted or sustained due to spacetime volumetric expansion as the universe expands ($H(t)$).

$-\frac{c^2}{a(t)^2}\nabla^2\psi$: Spatial propagation term; specifies the property of dark energy propagating in wave form along the expanding cosmic space ($a(t)$) rather than remaining stationary.

$m_{\mathrm{eff}}^2(\Phi)$: Effective mass/restoring force term; determined by the curvature of the potential $V(\Phi)$, providing elasticity (restoring force) that prevents the field from fluctuating excessively and drives it smoothly back to equilibrium.

$\mathcal{S}_{\mathrm{seq}}$: Sequence phase synchronization term on the RHS; a creative control term preventing dark energy from dispersing like random noise and causing it to undulate in a macroscopically coherent state with a smooth, uniform frequency (ω) and wave vector ($\mathbf{k}$). Thanks to this term, dark energy can be inferred as a gigantic "phase sequence wave" spanning the entire universe without local depletion.

$\mathbf{v}_{\mathrm{seq}}$: Vector field describing the spatial driving of the sequence (whether it represents actual fluid velocity or an effective driving field must be specified separately).

γ_{seq} : Coupling coefficient matching the dimension of the source term.

$\mathbf{k}$: Wave vector, with units of L^{-1}.

ω : Angular frequency, with units of T^{-1}.

Step 3: Unified Coherent Sequence Model: Incorporates self-interaction, phase driving, and dark matter coupling simultaneously:

$\mathcal{D}_0\delta\Phi = \mathcal{S}_{\rm nl}[\delta\Phi] + \mathcal{S}_{\rm drive} + \beta\rho_m$

($\mathcal{S}_{\rm nl}$ is the self-interaction term, $\mathcal{S}_{\rm drive}$ is the independently defined phase driving term, and any additional derivative coupling terms retained must be consistently reflected in $\mathcal{D}_0$ or the source term.) $\mathcal{S}_{\rm seq} = \mathcal{S}_{\rm nl} + \mathcal{S}_{\rm drive} + \mathcal{S}_K$

3. Higgs Field and Higgs Particles

Field Expansion, Mathematical Separation of the Higgs Field and Higgs Particles, and Higgs Field-Higgs Particle Wave Measurement Equation:

3.1. Higgs Field-Higgs Particle Wave Measurement Equation:

Combining the two split terms—the static mass generation term due to the vacuum expectation value (v), $\frac{g_f v}{\sqrt{2}}\bar{\psi}\psi$, and the dynamic interaction term due to Higgs field fluctuations ($H(x)$), $\frac{g_f}{\sqrt{2}}H(x)\bar{\psi}\psi$—the wave measurement equation is extended to simultaneously measure fermion matter waves and Higgs field fluctuations:

$$\left[i\gamma^\mu\partial_\mu - m_f - \frac{g_f}{\sqrt{2}}h(x)\right]\psi_f(x) = \eta_f(x)$$

$\eta_f(x)$: External spinor source directly coupled to the fermion equation. In natural units $\hbar = c = 1$, the left-hand and right-hand sides must have identical dimensions. To incorporate external probes such as electromagnetic pulses, the coupling of the electromagnetic field A_μ must be explicitly specified; for instance, minimal coupling for a charged fermion enters as $i\gamma^\mu\partial_\mu \longrightarrow i\gamma^\mu D_\mu$. Therefore, the measurement probe must not be unconditionally identified with η_f. The response for fermion scalar bilinears can be defined as an independent observable operator, $\mathcal{O}_f(x) = \bar{\psi}_f(x)\psi_f(x)$, and the linear response to an external probe $J(x)$ is expressed as:

$$\delta\langle\mathcal{O}_f(x)\rangle = \int d^4x'\,\chi^R_{\mathcal{O}\mathcal{O}}(x,x')J(x')$$

(where $\chi^R_{\mathcal{O}\mathcal{O}}$ is the retarded response function of the corresponding observable). This definition enables the clear separation of the measurement source from the physical quantity actually being measured.

Left-hand Operator: Indicates that the mass in the Dirac momentum term is not constant, but is dynamically modulated by the sum of the base mass field driven by the vacuum expectation value (v) and the spatially dependent Higgs fluctuation field ($h(x)$).

Right-hand Source Term ($\eta_f(x)$): An observational loss/response function designed to measure in real time the amplitude and phase variations of these two coupled waves via

externally induced probes (e.g., terahertz waves or high-frequency electromagnetic pulses).

3.2. Split-Component-Based Virtual Real-Space Measurement Amplitude Equation (Observable Wave Amplitude & Resonance Equation)

A complex amplitude superposition measurement model structured to separately measure the interference patterns of waves driven by the two split terms:

$$\Psi_{\text{total}} = \mathcal{A}_{\text{vac}}\,\mathcal{O}_{\text{vac}} e^{-i\Omega_{\text{mass}}t} + \mathcal{A}_{\text{fluct}}\,\mathcal{O}_{\text{H}} e^{i(\mathbf{k}_H\cdot\mathbf{x}-\omega_H t+\phi_{\text{mod}})}$$

$$\mathcal{O}_{\text{vac}} = \frac{g_f v_{\text{EW}}}{\sqrt{2}}\bar{\psi}_f\psi_f, \quad \mathcal{O}_{\text{H}} = \frac{g_f}{\sqrt{2}}h(x)\bar{\psi}_f\psi_f$$

Here, the complex amplitude synthesis expression $\Psi_{\text{total}}(x,t)$ itself is not directly regarded as a measurable physical quantity. Scattering cross-sections and response functions detected in actual laboratories and observations are defined via convolution rules with the retarded response function or the intensity $|\Psi_{\text{total}}(x,t)|^2$ of the fermion bilinear observable operator $\mathcal{O}_f(x) = \bar{\psi}_f(x)\psi_f(x)$.

Since both observables share identical dimensions, $\mathcal{A}_{\text{vac}}$ and $\mathcal{A}_{\text{fluct}}$ act as normalization coefficients. Both coefficients must be defined to properly match the dimension of the final amplitude Ψ_{total}. $\Omega_{\text{mass}} = \frac{m_f c^2}{\hbar}$:

Frequency corresponding to rest energy; the angular frequency of rest energy corresponding to mass m_f, representing the phase evolution of the stationary system for the first component. $I(x,t) \propto |\Psi_{\text{total}}(x,t)|^2$:

Rule for converting to a measurable quantity; a fundamental form for observing complex amplitude interference. Calculating actual scattering cross-sections requires additional definitions such as scattering amplitude normalization, incident states, and phase-space elements, but it is applied primarily as a measurement model representing amplitude interference.

Static Rest-Mass Wave Component (First Term): The intrinsic frequency oscillation component of fermions determined by the baseline VEV (v) of the Higgs sea, which establishes the particle's inertial mass scale.

Dynamic Fluctuation Wave Component (Second Term): Incorporates the spatial propagation and phase modulation (ϕ_{mod}) of the Higgs particle $h(x)$, enabling the capture of interference patterns and scattering cross-sections in real space when excitation states of the field resonate with matter waves.

Redefining these two terms as baseline mass background and spatially modulated phase waves allows us to analyze Higgs field fluctuations not merely as particle decay, but as macroscopic wave interference and phase-locking phenomena.

B. Master Unified Field System:

Macroscopic Sector: $\Phi = \overline{\Phi} + \delta\Phi, \quad \mathcal{D}_{\mathrm{DE}}\delta\Phi = \mathcal{J}_{\mathrm{DE}}$ ($\mathcal{D}_{\mathrm{DE}}$ is the dark energy perturbation operator, $\mathcal{J}_{\mathrm{DE}}$ is the source)

Microscopic Sector: $\phi = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_{\mathrm{EW}} + h \end{pmatrix}, \quad \mathcal{D}_{\mathrm{H}}h = \mathcal{J}_{\mathrm{H}}$ ($\mathcal{D}_{\mathrm{H}}$ is the Higgs fluctuation kinetic operator, $\mathcal{J}_{\mathrm{H}}$ is the source)

Isomorphic Mapping: $\mathcal{M}: (\overline{\Phi}, \delta\Phi, \mathcal{D}_{\mathrm{DE}}) \mapsto (v_{\mathrm{EW}}, h, \mathcal{D}_{\mathrm{H}})$ representing the correspondence between background and fluctuation fields, linear dynamical operators of each field, and sources/coupling terms. When operators and coupling terms do not strictly correspond, it denotes structural similarity or formal correspondence of background-fluctuation decomposition.

1. Comprehensive Equation Framework Consistently Connecting Macroscopic Spacetime Curvature to Microscopic Vacuum Fluctuations and Fermion Matter Waves

1.1. Macroscopic Background and Energy-Momentum Conservation (IDE & Dark Matter Framework)

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{(\mathrm{de})\mu\nu} + T_{(\mathrm{m})\mu\nu}\right)$$

$$\nabla_\mu T^{\mu\nu}_{(\mathrm{de})} = Q^\nu, \quad \nabla_\mu T^{\mu\nu}_{(\mathrm{m})} = -Q^\nu, \quad \nabla_\mu\left(T^{\mu\nu}_{(\mathrm{de})} + T^{\mu\nu}_{(\mathrm{m})}\right) = 0$$

$$\rho_m(a) = \rho_{m0}a^{-3}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P), \quad v(r) = \sqrt{\frac{GM(r)}{r}}$$

1.2. Macro-Micro Isomorphic Field Decomposition

$$\Phi(\mathbf{x}, t) = \Phi_0(t) + \psi(\mathbf{x}, t), \quad \phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_{\mathrm{EW}} + H(x) \end{pmatrix}$$

1.3. Coherent Sequence Wave Equation Sector (Macro Dark Energy & Matter Coupling Wave)

$$\left[\frac{\partial^2}{\partial t^2} + 3H(t)\frac{\partial}{\partial t} - \frac{c^2}{a(t)^2}\nabla^2 + m_{\mathrm{eff}}^2(\Phi_0)\right]\psi(\mathbf{x}, t) = \mathcal{S}_{\mathrm{seq}}[\psi] + \beta\rho_m(\mathbf{x}, t)$$

1.4. Micro-Vacuum Higgs and Yukawa Coupling Measurement Wave Sector

$$V(\phi) = -\mu^2\left(\phi^\dagger\phi\right) + \lambda\left(\phi^\dagger\phi\right)^2, \quad v^2 = \frac{\mu^2}{\lambda}, \quad (\mu^2 > 0, \lambda > 0)$$

Higgs doublet expansion:

$$\phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + H(x) \end{pmatrix}$$

$$\left[i\gamma^{\mu}\partial_{\mu} - \left(\frac{g_f v}{\sqrt{2}} + \frac{g_f}{\sqrt{2}}H(x)\right)\right]\psi_f(x) = \mathcal{S}_{\text{meas}}(x,t)$$

$$m_f = \frac{g_f v}{\sqrt{2}}, \quad \left[i\gamma^{\mu}\partial_{\mu} - m_f - \frac{g_f}{\sqrt{2}}H(x)\right]\psi_f(x) = \mathcal{S}_{\text{meas}}(x,t)$$

Complex amplitude superposition equation:

$$\Psi_{\text{total}}(x,t) = \underbrace{\mathcal{A}_{\text{vac}}\left(\frac{g_f v}{\sqrt{2}}\bar{\psi}_f\psi_f\right)e^{-i\Omega_{\text{mass}}t}}_{\text{Static Rest-Mass Wave}} + \underbrace{\mathcal{A}_{\text{fluct}}\left(\frac{g_f}{\sqrt{2}}H(x)\bar{\psi}_f\psi_f\right)e^{i(\mathbf{k}_H\cdot\mathbf{x}-\omega_H t+\phi_{\text{mod}})}}_{\text{Dynamic Higgs Excitation Wave}}$$

2. Nomenclature and Functional Descriptions of All Symbols in the Equations

2.1. Spacetime, Macroscopic Geometry, and Micro-Vacuum Higgs & Yukawa Symbols:

$g_{\mu\nu}$: Spacetime metric tensor, defines geometrical distances and spacetime curvature.

$R_{\mu\nu}$: Ricci tensor, degree of spacetime distortion.

R : Scalar curvature, total sum of spacetime curvature.

Λ : Cosmological constant, vacuum energy density.

G : Gravitational constant, determines the strength of gravity.

c : Speed of light.

$a(t)$: Scale factor, rate of spatial size variation due to cosmic expansion.

$H_{\text{Hub}}(t)$: Hubble parameter ($\frac{\dot{a}}{a}$), expansion rate of the universe.

λ_H : Higgs potential self-coupling constant.

2.2. Dark Energy and Dark Matter Symbols (IDE & Dark Sector):

$T^{\mu\nu}_{(\text{de})}$: Dark energy energy-momentum tensor.

$T^{\mu\nu}_{(\text{m})}$: Dark matter energy-momentum tensor.

Q^{ν} : Energy-momentum transfer vector, controls interactions between the two components.

$\rho_m(a)$: Dark matter density as a function of cosmic expansion.

ρ_{m0} : Reference dark matter density at the present epoch.

ρ_{DM} or $\varepsilon_{\mathrm{DM}}$: Mass density or energy density.

$v_{\mathrm{rot}}(r)$: Galaxy rotation velocity at radius r.

$M(r)$: Enclosed total mass within radius r.

λ_{DE} : Dark energy Lagrangian coupling constant.

ψ_{DE} : Macroscopic fluctuation field.

2.3. Coherent Sequence Wave Equation Symbols:

$\delta\Phi$: Dark energy perturbation; local variation from the background.

Φ_{DE} : Macroscopic dark energy field.

$\Phi(\mathbf{x}, t)$: Total macroscopic scalar field.

$\Phi_0(t)$: Background scalar field, average background of dark energy.

$\psi(\mathbf{x}, t)$: Fluctuation wave function, local undulations of the field.

$m_{\mathrm{eff}}^2(\Phi_0)$: Effective mass squared, restoring force frequency scale of the field (T^{-2}).

m_{eff}^2 or Ω_{eff}^2 : Effective mass squared or frequency squared (T^{-2} in standard canonical field equations).

$\mathcal{S}_{\mathrm{seq}}[\psi]$: Self-interaction sequence source term, controls phase synchronization within the field.

$\mathcal{S}_{\mathrm{nl}} + \mathcal{S}_{\mathrm{drive}} + \mathcal{S}_K$: Corrections for self-interaction, driving, and kinetic terms.

β : Matter coupling constant, defined such that $\beta\rho_m$ shares the same dimension as the LHS operator; controls the impact of dark matter density on the field.

∇^2 : Three-dimensional spatial Laplacian differential operator.

2.4. Micro-Vacuum Higgs and Fermion Symbols:

$\phi(x), \phi_{\mathrm{H}}$: Electroweak Higgs doublet field.

$V(\phi)$: Mexican-hat Higgs potential.

μ^2, λ : Higgs potential parameters ($\mu^2 > 0, \lambda > 0$).

λ_{seq} : Additional derivative coupling for dark energy.

λ_{H} : Higgs potential coupling.

v_{EW} : Vacuum expectation value (VEV).

$h(x)$: Higgs particle fluctuation field.

γ^{μ} : Dirac matrices, describes relativistic motion of spin-1/2 fermions.

$\psi_f(x)$: Fermion matter wave field (e.g., electrons, quarks).

g_f : Yukawa coupling constant, interaction strength between fermions and the Higgs field.

$\mathcal{S}_{\mathrm{meas}}(x,t)$: Measurement probe source function.

$\Psi_{\mathrm{total}}(x,t)$: Complex amplitude superposition wave function of static mass waves and dynamic Higgs waves.

Ψ_{total} or $\mathcal{A}_{\mathrm{total}}$: Composite variable of complex amplitude; model variable representing interference between the two components (requires distinction as to whether this variable represents the fermion field itself, complex amplitude, or a detectable observable).

$\mathcal{A}_{\mathrm{vac}}, \mathcal{A}_{\mathrm{fluct}}$: Amplitude scales for each wave component.

$\Omega_{\mathrm{mass}}, \omega_H$: Static mass oscillation frequency and Higgs fluctuation frequency, respectively.

$\mathbf{k}_H$: Higgs fluctuation wave vector.

ϕ_{mod} or θ_{mod} : Dimensionless phase modulation angle.

Φ : Dark energy background field; homogeneous background solution.

$\mathcal{S}_{\mathrm{nl}}$: Nonlinear self-interaction term; higher-order potential terms, etc.

$\mathcal{S}_{\mathrm{drive}}$: External phase driving term; spatial and temporal driving of sequences.

η_f or J_{probe} : Distinction between external fermion spinor source and external measurement probe (RHS of the Dirac equation).

$\mathcal{O}_f$: Fermion observable operator; e.g., $\bar{\psi}_f \psi_f$.

ω_{H} : Higgs fluctuation angular frequency.

m_{eff}^2 must be carefully distinguished from actual mass squared. If actual mass M_{eff} is used, the term is expressed as $\frac{M_{\mathrm{eff}}^2 c^4}{\hbar^2}$; if m_{eff} is defined as an effective angular frequency, the dimension of m_{eff}^2 is T^{-2}.

ρ_{DM} $[\mathrm{kg\,m^{-3}}]$: Mass density.

$\varepsilon_{\mathrm{DM}} = \rho_{\mathrm{DM}} c^2$ $[\mathrm{J\,m^{-3}}]$: Energy density.

When utilizing $\beta\rho_{\mathrm{DM}}$ as a source term in field equations, the dimension of β must be defined. If energy density is used as the source, the coupling constant must be redefined accordingly.

Unification of sequence source terms: $\mathcal{S}_{\mathrm{seq}} = \mathcal{S}_{\mathrm{nl}} + \mathcal{S}_K + \mathcal{S}_{\mathrm{drive}}$

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